* **Testing of Hypothesis:**

- Parametric Test — Test Based on Statistical Assumptions
  - Types:
    - Small Sample Test : (Sample Size less than 30)
      - Arthemetic Mean — t-test
        - Independent
          - Sample mean with Hypothetical mean
          - Equality of two Sample mean
        - Dependent
          - Paired t-test for Arthmetic mean
      - Variance
        - F-test
          - Homogeneity of Sample or Equality of two variances of Small Sample
          - Equality of more than 2 variances (Bartlett's Test)
    - large Sample Test
- Non-parametric Test
  - No Statistical Assumptions
  - Chi-Square Test
  - Wilcoxon Signed Rank Test
  - Wilcoxon Matched Pair Rank Test
  - Sign Test
  - Run Test
  - Mann Whitney U Test
  - Kruskal Wallis Test
  - Fried Man Rank Test
  - Additional Test for Quantitative

Test Statistic $= \dfrac{\text{Statistic} - \text{Parameter}}{SE(\text{Statistic})} \sim$ Prob distribution with df and α %.

* **Chi-Square Test:**
  - Independence of Attributes
  - Goodness of fit

* One-Sample t-test : $t = \frac{|\bar{x} - \mu|}{SE(\bar{x})} \sim t_{n-1}$ df, α %

$\bar{x}$ → sample mean

α % → Significance level

$n-1$ → degree of freedom

$SE(\bar{x})$ → Standard error

H_0 : observed mean = Hypothesized mean

H_1 : H_0 is not true

$\therefore SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$

* Dependent Sample t-test

$t = \frac{\bar{d}}{SE(\bar{d})} \sim t_{n-1}$ df, α %

where, $\bar{d}$ = difference

$SE(\bar{d}) = \frac{\sigma_d}{\sqrt{n}}$

* Two Sample Independent t-test : $t = \frac{|\bar{x}_1 - \bar{x}_2|}{SE(\bar{x}_1 - \bar{x}_2)} \sim t_{n_1+n_2-2}$ df 'α %

n_1+n_2-2 → degree of freedom

Samples from Homogenous population

$= \sqrt{S^2\left[\frac{1}{n_1-1} + \frac{1}{n_2-1}\right]}$

S^2 → combined or pooled variance

$= \frac{S_1^2(n_1-1) + S_2^2(n_2-1)}{n_1+n_2-2}$

Samples from Heterogenous population

$= \sqrt{\frac{S_1^2}{n_1-1} + \frac{S_2^2}{n_2-1}}$

→ check using Test of Homogeneity

F-Test → H_0 : Homo, H_1 : Hetero

Check $S_1^2 \times S_2^2$:

- $F = \frac{S_1^2}{S_2^2} \sim F_{((n_1-1),(n_2-1))}$ df, α % if $S_1^2 > S_2^2$
- $F = \frac{S_2^2}{S_1^2} \sim F_{((n_2-1),(n_1-1))}$ df, α % if $S_2^2 > S_1^2$

Large Sample Test

1 sample

$$Z = \frac{|\bar{x} - \mu|}{SE(\bar{x})} \sim N_{\mu, \sigma^2, \alpha\%}$$

Table value : 5% → 1.96
1% → 2.576

2 sample

* Z-test:

$$Z = \frac{|\bar{x}_1 - \bar{x}_2|}{SE(\bar{x}_1 - \bar{x}_2)} \sim N_{\mu, \sigma^2, \alpha\%}$$

where,

$$SE(\bar{x}_1 - \bar{x}_2) = \sqrt{S^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$S^2 = \frac{S_1^2(n_1) + S_2^2(n_2)}{n_1 + n_2}$$

* Z-test (Proportion Test)

- Single proportion with Hypothetical proportion → $Z = \frac{|p - P|}{SE(p)} \sim N(0,1)$

where,

$$SE(p) = \sqrt{\frac{PQ}{n}} \; ; \; Q = 1 - P$$

- Two Sample proportion equality

$$Z = \frac{|p_1 - p_2|}{SE(p_1 - p_2)} \sim N_{\mu, \sigma^2, \alpha\%}$$

where,

$$SE(p_1 - p_2) = \sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$p = \frac{p_1 n_1 + p_2 n_2}{n_1 + n_2} \; , \; q = 1 - p$$

* Chi-Square Test:

- Independence of Attributes:

H_0: Attributes are independent

H_1: Attributes are dependent.

① $\chi^2 = \dfrac{(ab-bc)^2 N}{R_1 \cdot R_2 \cdot C_1 \cdot C_2} \sim \chi^2_{1\ df, \alpha\%}$

Note: if 2 rows & 2 columns then,

$df = (r-1) * (c-1)$
$= (2-1) * (2-1)$
$= 1$

- Contingency Table:

	A1	B1	
A2	a	b	R1
B2	c	d	R2
	C1	C2	$\boxed{N}$

② $\chi^2 = \sum\left[\dfrac{(O_i - E_i)^2}{E_i}\right] \sim \chi^2_{\text{calculated df}, \alpha\%}$

* ANOVA (Analysis of Variance): [Notes on one-factor ANOVA]

- One-way ANOVA (controlled condition) → Treatment
- Two-way ANOVA → Treatment + Replication
- $\boxed{\text{Factorial ANOVA}}$ → Row + column + Treatment

(one factor ANOVA)

- Equality of more than two sample means
- Introduced & done by Ronald. A. Fisher
- uses F-test to give conclusion of experimental Results
- Deals with experimental data not observational data

Note:

Experimental data are observed ~~at~~ controlled conditions.

* **Assumptions of ANOVA:**

- Factors in general linear model are linear in nature or additive in nature
- All the population in analysis Having common Variance
- Within each group, Samples are drawn randomly, and normally distributed with mean μ & common variance σ^2.
- All Samples which are drawn, Should be independent with each others.
- Error term in model Should be normally distributed with mean 0 and common variance σ^2

$\therefore$ Error (observed - expected) $\sim N(0, \sigma^2)$

one-way ANOVA model:

$$y_{ij} = \mu + T_i + \varepsilon_{ij}$$

y_{ij} → Observation of Treatment.

μ → Grand mean

T_i → effect of Treatment [Difference of Group mean with overall mean]

ε_{ij} → error of observation.

* choosing Statistical Test.

1. Type of Variable:
   - Numerical
     - continuous
     - Discrete
   - Categorical

2. Type of Analysis:
   - comparision
     - mean
     - median
     - proportion
   - Relation
   - Prediction

3. No. of groups:.
   - one → 2 dataset; → more than 2 dataset
   - Two → 2 dataset
   - More than 2 → more than 2 dataset

4. Study Design:
   - unpaired / Independent
   - paired / Matched

5. Distribution of Data:
   - Normal
   - Non-normal
   - Dichotomous

	Comparision				Association [correlation between 2 variable]	Regression [Predicting one from another]
	2 datasets		>2 datasets			
	Paired	unpaired	paired	unpaired		
Normal Distribution (Mean)	Paired t-test	unpaired t-test	Repeated Measures ANOVA	one way ANOVA	Pearson Correlation	Linear Regression
Non-Normal Distribution (Median)	Wilcoxon Signed Rank Test	Wilcoxon Rank Sum Test Mann Whitney U Test	Friedman Test	Kruskal Wallis Test	Spearman's Rank Correlation	Non-parametric Regression
Dichotomous Data	McNemar's Test	chi-square Test Fisher's exact Test	Cochran's Q Test	chi-square Test	Contingency Coefficient	Logistic Regression